



Similarity design and experimental investigation of a beta-type Stirling engine with a rhombic drive mechanism

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SUMMARY

Stirling engines are power machines that operate over a closed, regenerative thermodynamic cycle with the ability to use any heat source from the outside, including hydrogen, solar energy, and biomass fuels. In this work, the development of a beta-type Stirling engine is presented. The improved similarity design and optimization methods are described in detail, as are the key parameters of the constructed prototype and the arrangement of the entire test rig. A new structure for the expansion exchangers is developed to reduce the flow loss. The performance test of the prototype engine is conducted under laboratory conditions using an electrical heating system. In this test, the temperature and the pressure of the working fluid are monitored by thermocouples and pressure sensors, respectively. The speed and the torque of the output shaft are obtained by the dynamometer. Finally, the preliminary test results with the prototype engine are shown. The maximum output shaft power can reach 288 W at 600°C and 15-bar charge pressure. Copyright © 2014 John Wiley & Sons, Ltd.

KEY WORDS

Stirling engine; beta type; rhombic drive; similarity design; charge pressure; engine performance

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1. INTRODUCTION

The increasing global concerns related to energy conservation and environmental protection have resulted in major efforts to improve the uptake of alternative energy resources and the efficiency of the utilization of the conventional carbon-based fuels [1]. Recently, extensive research has been conducted to advance the technologies of distributed power generation systems based on various renewable energy sources, such as solar energy, geothermal, biomass, and industrial waste heat, because the use of renewable heat resource has a beneficial effect on greenhouse gas emissions because of the saving in additional carbon-based fuels. Several competing technologies can be used for such power generation: solar, geothermal, or waste heat-based Organic Rankine cycle [2–4], Stirling engine [5,6], and hybrid fuel cell power systems [7,8]. In particular, the Stirling engine has been regarded as one of the most promising sustainable energy technologies in recent years [9,10]. It is also a good option for a remote district as a distributed energy system.

The Stirling cycle has its origins in 1816, being first described in a patent granted in that year to Robert Stirling [11]. The Stirling engine is based on simple concepts. In

fact, the Stirling cycle engine was conceived more than 80 years before the diesel and the Otto cycle engines. Since its conception, the Stirling engine has been recognized for its potential to attain high cycle efficiency; however, it has failed to reach its full potential because of design challenges such as seals, materials, heat transfer, size, and weight. In the past 200 years, individual inventors and corporate research laboratories have produced scores of technical and patent literature reports on thousands of Stirling engine designs.

The Stirling engine is a reciprocating, externally heated engine [12]. Heat is transferred to the working gas and is then converted to work by expanding the gas inside the cylinder. Heat is added to the expanding gas continuously and at a constant temperature. This constant temperature heat addition implies a very high heat transfer rate through the cylinder, which is one of the main technical challenges with the Stirling engine. The conventional, reciprocating internal combustion (IC) engine, by comparison, burns an air–fuel mixture inside the cylinder to generate the heat and the pressure, which is converted to work at the crankshaft. The temperature is generally not constant during the power stroke in the IC engine, but it varies as the combustion and the piston motion proceed. Because

heat is supplied externally to the Stirling engine, a wide variety of heat sources can be used: heat of combustion of any fuel such as hydrogen or biogas, solar irradiation, heat from radioisotopes, and waste heat from industrial processes [13].

Stirling engines can be broadly classified into three types of configurations [12] as follows:

1. Alpha configuration: There are two or more separate pistons in Alpha engines that are mechanically linked to oscillate with a constant phase lag.
2. Beta configuration: Beta engines have a displacer–piston arrangement in which the displacer and the power piston are in line with each other.
3. Gamma configuration: Gamma engines have a displacer–piston arrangement where the displacer and power pistons are in separate cylinders.

A representative example of each of these three arrangements is shown in Figure 1. Among these three arrangements, the beta-type engine has higher power density than the other two types. Therefore, it is chosen in this study to build a prototype engine.

The rhombic drive mechanism, a special drive mechanism for the beta-type Stirling engine, was invented by R.J. Meijer of Philips in 1959 [14] and is shown schematically in Figure 2. The system consists of separate displacer and power pistons, which have conveniently separate functions. The displacer separates the hot and cold spaces at approximately the same pressure and is used to shuttle the working gas between these spaces. The power piston provides both the compression and the expansion spaces. There are two cranks, which are geared together to counterrotate and drive the yokes, to which the displacer and the power piston rods are attached. A pressured buffer space allows the engine to operate at high mean pressures without requiring a pressured crankcase. Thus, only two shaft seals are required on the displacer and the power piston rods. The success of this arrangement is because the two pistons have no side forces, as the horizontal

components of the forces exerted by each pair of connecting rods are exactly balanced at each yoke. Furthermore, complete static and dynamic balancing is attainable in the single-cylinder arrangement, allowing vibrationless high-speed operation. The principal disadvantage of the rhombic drive engine is its relative complexity. Because it has been proven to be a good structure for Stirling engines, much work has been performed on beta-type Stirling engines, especially in the form of simulations [15–17]. However, so far, study on the performance of the rhombic drive Stirling engine is very limited.

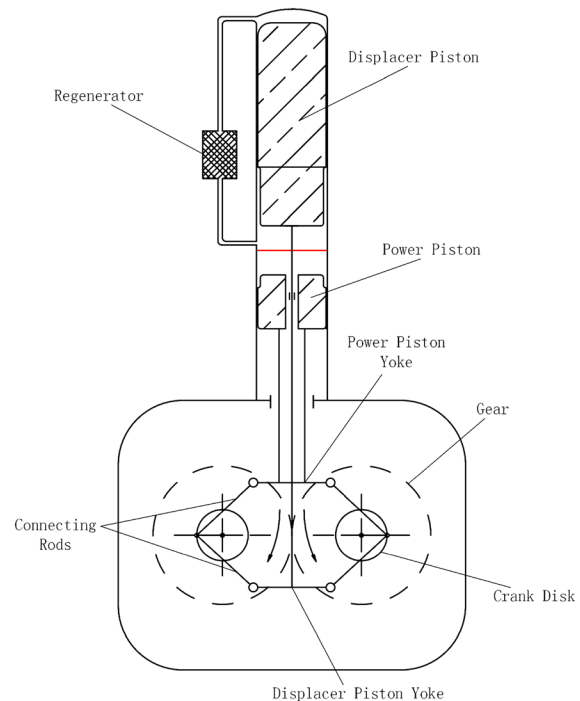
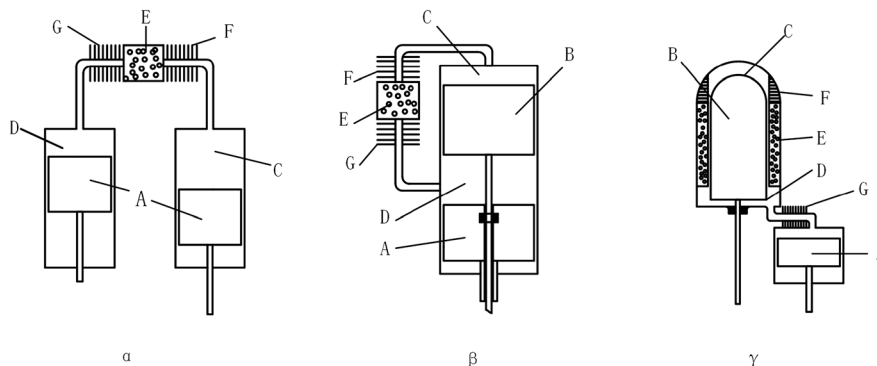


Figure 2. Schematic of the rhombic drive Stirling engine.



A – piston; B – displacer; C – expansion space; D – compression space; E – regenerator; F – heater; G – cooler

Figure 1. Three basic arrangements of Stirling engines.

The invention of the Stirling engine preceded modern thermodynamic theory to a degree exemplified by the fact that it was not until 50 years after its invention that a reasonable analysis of the cycle was published. This analysis was made by Gustav Schmidt in 1871 [18] and has been reported in a more comprehensive form in the book of Israel Urieli [19]. The analysis, however, is limited in that it assumes isothermal working spaces and ideal heat exchangers, and thus, it calculates the efficiency as being only temperature dependent (i.e., Carnot efficiency). In 1960, Finkelstein presented an ideal analysis, in which he assumed adiabatic working spaces but maintained the heat exchangers as ideal [20]. The effects of heat transfer and imperfect regeneration on the performance of the irreversible Stirling engine were investigated by Wu *et al.* [21]. As efforts to reduce certain loss mechanisms can tend to increase the detrimental impact of others, complementary approaches are needed. It is noted that a practical Stirling cycle always departs significantly from the ideal cycle, mainly because of the performance of the heat exchanger and the regenerator. A detailed analysis with practical values and examples of the engine were given by Organ [22]. The similarity and scaling approach developed by Organ provides a simpler way of creating the preliminary design for the prototype engine. In 2000, Kaushik and Kumar presented an investigation of a finite-time thermodynamic analysis of a Stirling engine in which the power output and the corresponding thermal efficiency was maximized [23]. Petrescu and Costea [24] conducted a technique for calculating the efficiency and power of Stirling engines based on the first law of thermodynamics for processes with finite speed [24,25]. In 2008, Timoumi and Tlili developed a second-order Stirling model that includes thermal losses and is applied for the optimization of generator protection unit (GPU)-3 Stirling engines [26]. Leonardo Scollo [27] presented an alpha-type Stirling engine by using a similar design method. A numerical model for the beta-type Stirling engine with a rhombic drive mechanism was developed by Cheng *et al.* [28]. The energy equations for the control volumes in the working chambers and heat exchangers were derived and solved by taking into account the nonisothermal effects, the thermal resistance of the heater head, and the effectiveness of the regenerative channel [28]. Formosa *et al.* [29] developed an analytical model that takes into account the heat losses and irreversibility on Stirling engines that could be used in a dynamical analysis for preliminary design [29]. In 2011, a semianalytical dynamic model of a free piston Stirling engine was developed by Formosa [30]. The thermodynamic model is used to define the thermal variables that are used in a dynamic model that evaluates the kinematic results. Fethi Aloui *et al.* [31] used the experimental design method to identify the parameter sets under which the Stirling engine can obtain optimal performance. Recently, Aksoy and Cinar [32] conducted a theoretical investigation on dynamic and thermodynamic analysis of a beta-type Stirling engine with a rhombic drive mechanism by implementing a nodal analysis method [32].

Although there are a lot of published studies about the numerical analysis of Stirling engines, the development of a prototype engine is very limited especially the type of rhombic drive Stirling engine. In the present study, a beta-type Stirling engine with the rhombic drive mechanism was designed and constructed. A new structure for the expansion exchangers is developed to reduce the flow loss. A modified structural design method and the similarity approach were used during the design process. The performance tests of the prototype engine were carried out at different charge pressures and hot end temperatures.

2. DESIGN OF THE ENGINE PROTOTYPE

2.1. Design of the rhombic drive mechanism

Analysis of the rhombic drive includes the determination of stroke, phase angle, crank length r , connecting rod length L , and eccentricity e . The stroke of the piston is equal to that of the displacer because of the symmetry of the rhombic drive about the y -axis, as shown in Figure 2. The definition of terms of the mechanism is described in Figure 3. The displacer is in its lowest position when the displacer connecting rod and the crank are in a straight line, and it is in its top position when the displacer connecting rod covers the crank [33]. It follows that

$$\frac{S}{r} = \frac{1}{\lambda} \left(\sqrt{(1+\lambda)^2 - \lambda^2 \varepsilon^2} - \sqrt{(1-\lambda)^2 - \lambda^2 \varepsilon^2} \right) \quad (1)$$

where the eccentricity ratio is $\varepsilon = e/r$ ($\varepsilon > 1$) and $\lambda = r/L$, $(e+r) < L$. Thus, the stroke can be calculated by using a simple relation with the variables ε and λ .

The rhombic drive mechanism does not produce sinusoidal volume variations. The actual volume variations are functions of the various geometric parameters of the

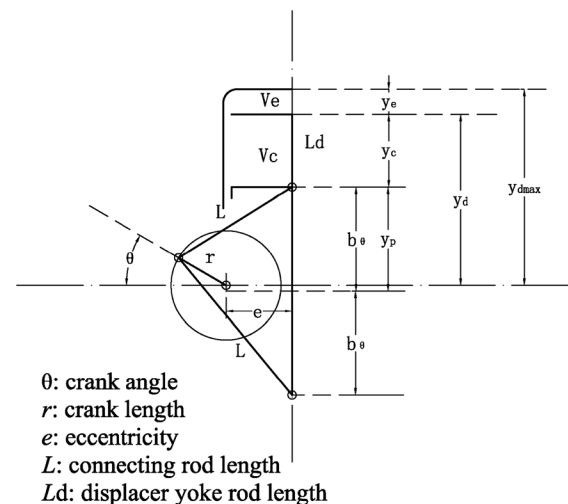


Figure 3. Terminology for analysis of the rhombic drive mechanism.

mechanism and could be derived from geometrical considerations. As depicted in Figure 4, there are four specific positions during the operation processes: (i) When the crank angle equals 0° , the engine reaches the maximum compression volume; (ii) when the rod connected with the displacer yoke covers the crank radius, the engine reaches the minimum expansion volume; (iii) when the crank angle is π , the engine reaches the minimum compression volume; and (iv) when the rod connected with the displacer yoke keeps in the extension cord of the crank radius, the engine reaches the maximum expansion volume. Consequently, the displacement equations of the rhombic drive mechanism can be expressed as follows:

- Distance between the top surface of the power piston and the center line

$$y_p = r \sin \theta + b_\theta \quad (2)$$

- Distance between the top surface of the displacer and the center line

$$y_d = L_d - b_\theta + r \sin\theta \quad (3)$$

- ### 3. Distance between the displacer and power piston

$$y_c = L_d - 2b_\theta \quad (4)$$

4. Distance between the displacer and top dead center

$$y_e = y_{d \max} - y_d \quad (5)$$

where $b_\theta = \sqrt{L^2 - (e + r \cos \theta)^2}$.

Furthermore, the geometric derivation of the rhombic drive equations can also be deduced as follows:

- ### 1. Volume variation of compression space

$$V_c = V_{clc} + 2A_p(b_1 - b_\theta) \quad (6)$$

- ## 2. Volume variation of expansion space

$$V_e = V_{clc} + A_d(b_\theta - b_2 - r \sin\theta) \quad (7)$$

- ### 3. Variation of power piston swept volume

$$V_{swc} = 2A_p(b_1 - b_3) \quad (8)$$

- #### 4. Variation of displacer swept volume

$$V_{swe} = A_d(b_4 - b_2) \quad (9)$$

From the equations mentioned previously, the stroke of the displacer and the piston can be described in Figure 5. Some key design parameters of the rhombic drive mechanism

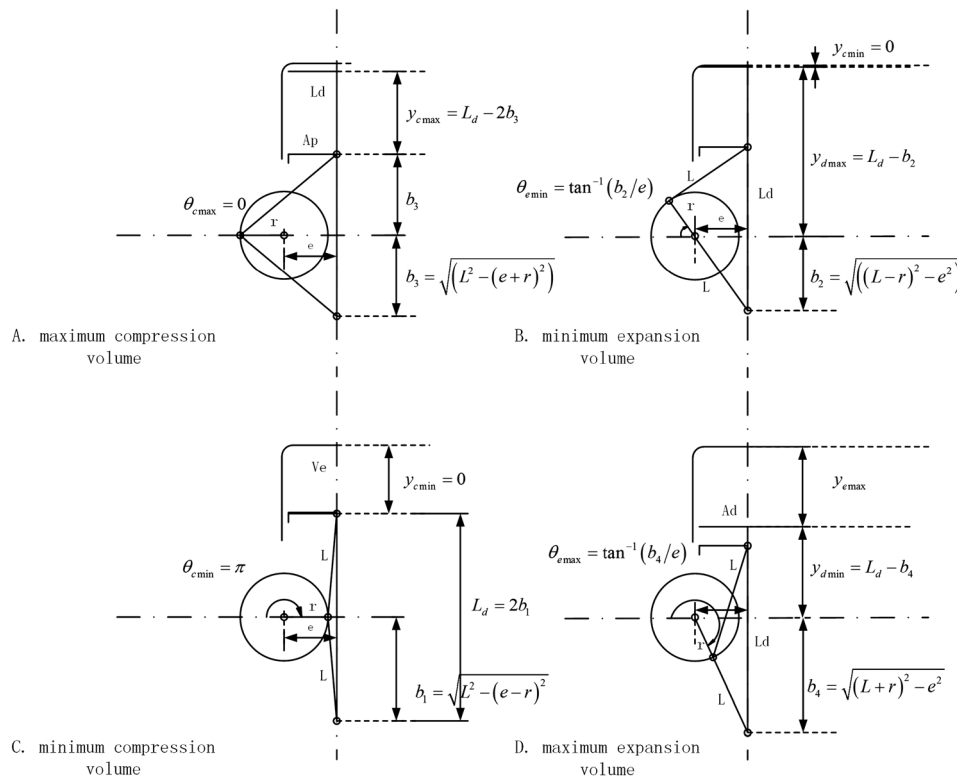


Figure 4. Four specific positions of the rhombic drive mechanism.

are shown in Table I. The rhombic drive components are illustrated in Figure 6.

2.2. Similarity design of the engine prototype

The Stirling cycle machine works by virtue of convective heat transfer and in the process suffers from flow losses.

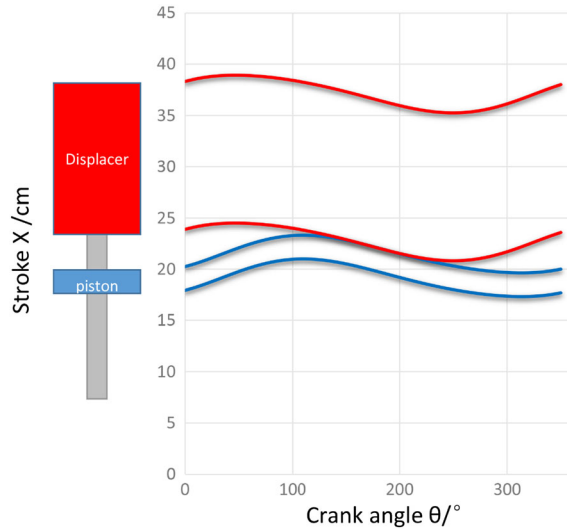


Figure 5. Stroke variations of the displacer and piston.

Table I. Key parameters of the rhombic drive mechanism.

Parameter	Value
Connecting rod length	4.80 cm
Crank radius	1.60 cm
Eccentricity	2.20 cm
Displacer/piston yoke length	4.00 cm
Swept area	21.64 cm ²
Displacer stroke	3.69 cm
Piston stroke	3.66 cm
Phase angle advance	121°

To this extent, it is more than a compound heat exchanger subject to cyclically varying pressure. Its characteristics are therefore susceptible to correlation by the same principles as those of the isolated exchanger. The correlation is a basis for scaling and, in turn, for thermodynamic design.

In a Stirling engine, heat exchanger dimensions do not generally scale linearly with size nor is there any reason why they should. However, actually, they do scale, so it is necessary to establish the rules governing the scaling process. These are the rules that preserve the thermodynamic processes when size, speed, working fluid, charge pressure, etc. are changed. Scaling is an approach to thermodynamic design whereby the (unknown) gas processes of an engine of known performance may be replicated in a new gas circuit of different specification.

As depicted in Figure 7, Finkelstein's elegant reworking of the Schmidt analysis provides an example [20]. The basic concept amounts simply to a statement of the mass content of each space of the gas circuit in terms of the ideal gas equation. The sum is equal to total mass, M , taking part in the cycle. Inversion of the expression for M yields instantaneous cycle pressure. The treatment supposes simple-harmonic variation of the volumes of expansion and compression spaces with a constant volume phase angle, α

$$V_e(\theta) = \frac{1}{2} V_E (1 + \cos(\theta)) \quad (10)$$

$$V_c(\theta) = \frac{1}{2} V_C (1 + \cos(\theta - \alpha)) \quad (11)$$

Volume V_i at uniform temperature T_i has an instantaneous mass content m_i at pressure p given by $m_i = pV_i/RT_i$. Summing the constituent m_i ,

$$M = \frac{pV_E}{RT_C} \left(\frac{1}{2} (1 + \cos(\theta)) T_C/T_E + \frac{1}{2} (V_C/V_E) (1 + \cos(\theta - \alpha)) + \sum V_d T_C/V_E T_i \right) \quad (12)$$

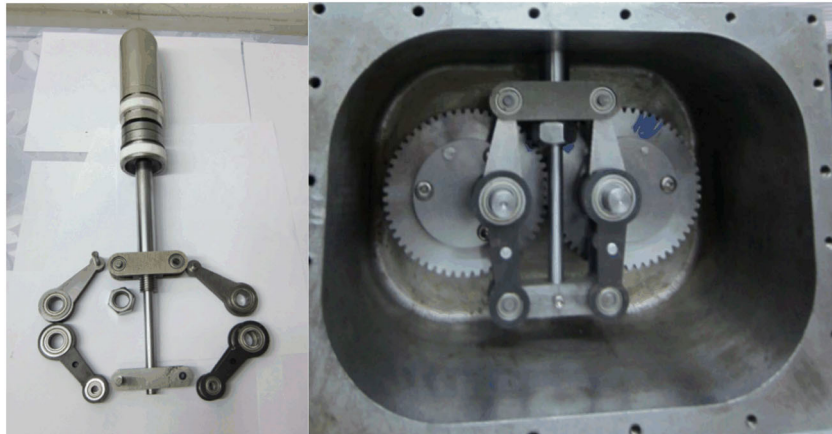


Figure 6. Components of the rhombic drive mechanism.

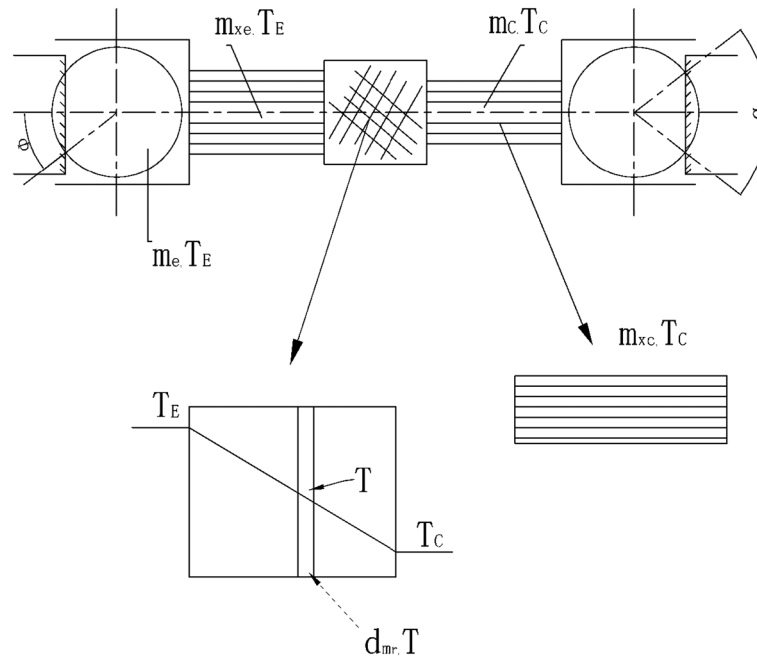


Figure 7. Stirling cycle machine as envisaged in Schmidt analysis.

The dimensionless pressure Ψ has been defined as pV_E/MRT_C , then

$$\Psi = \frac{1}{\frac{1}{2}((1 + \cos(\theta))/N_T + \kappa(1 + \cos(\theta - \alpha)) + 2v(N_T))} \quad (13)$$

where Ψ is a function of five parameters temperature ratio $N_T = T_E/T_C$, volume ratio $\kappa = V_C/V_E$, volume phase angle α , dead space parameter $v(N_T) = \sum V_i T_C / V_E T_i$, and crank angle θ

$$\psi = \psi(N_T, \kappa, \alpha, v, \theta)$$

It is possible to proceed in terms of the ratio $\psi/\psi_{\max} = p/p_{\max}$, in which case the parameters combine and are reduced to three, namely $(N_T\kappa)$, α , and (N_Tv) . Each engine having the same values of these three parameters has the same variation of pressure ratio with crank angle θ .

In the present study, similarity and the scaling method were adopted for the engine design. The design method mentioned before was first described by Organ [22]. The similarity of computed thermodynamic performance between two Stirling engines depends on the corresponding working fluid particles that undergo similar processes at corresponding points in the cycle. In the first step, the original Stirling engine was chosen, from which the prototype engine was scaled. The GPU-3 Stirling engine [19], which was developed by General Motors Research Laboratories, was chosen as the model because of its similar geometry and arrangement. The operating point and the dimensionless parameters of the GPU-3 are shown in Table II. The scaled design results obtained from the mentioned method are shown in Table III.

Table II. Operating point, dimensionless gas circuit description, and similarity parameter value for GPU-3.

Parameter		Values
Operating points	Working fluid	He
	P_{ref}	70 bar
	Speed	3600 rpm
	Power	8.95 kW
	V_{sw}	118 cm ³
	S	48.8 mm
	λ_{xe}	5.02
	λ_r	0.46
	λ_{xc}	0.94
	α_{fxe}	0.12
Geometric	α_{ffr}	1.00
	α_{ffxc}	1.19E-01
	λ_{hxe}	1.54E-02
	λ_{hr}	6.10E-04
	λ_{hxc}	5.50E-03
	Π_V	0.75
	N_T	3.28
	N_{MA}	1.66E-02
	N_{SG}	2.18E+09
	N_{TCR}	193
Working condition	N_F	6.05

3. EXPERIMENTAL SETUP AND TEST PROCEDURE

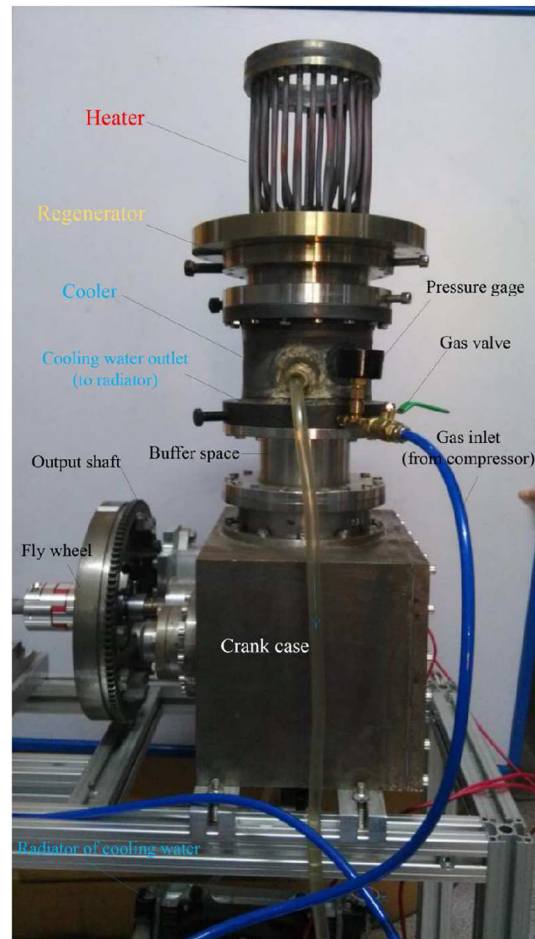
The parallel-bore, coaxial, beta-type Stirling engine with a rhombic drive mechanism developed by the authors is depicted in Figure 8. The heater, the regenerator, and the cooler all have an annular configuration and are mounted

Table III. Technical specification of the engine prototype.

	Parameter	Value
Heater	Number of heater tubes	28
	Heater tube length	16.00 cm
	Tube inside diameter	0.30 cm
	Tube outside diameter	0.45 cm
	Designed wall temperature	600°C
Cooler	Number of cooler tubes	38
	Cooler tube length	10.00 cm
	Tube inside diameter	0.15 cm
	Tube outside diameter	0.25 cm
	Designed wall temperature	20°C
Regenerator	Diameter (inside)	8.97 cm
	Diameter (outside)	10.37 cm
	Matrix material	Stainless steel
		wire cloth
	Number of wires, per inch	200 × 200
	Wire diameter	0.035 mm
	Filler factor, percent	30.1
	Angle of rotation between adjacent screens	5°
	Working fluid	Air
	Heating method	Electrical heater
Working state	Mean pressure	1 MPa
	Designed speed	800 rpm

around the expansion–compression cylinder. This arrangement allows simple regenerator dismounting for tests of different regenerator matrices. The heater consists of two sets of tubes, arranged alternately with a small clearance, as illustrated in Figure 9. These two sets of tubes are connected by a common channel settled on the upper head of the tubes. The tubes of the one set terminate in the regenerator inlet; the gas coming from the regenerator flows up these tubes and via the common annular channel into the downwardly directed tubes, which are soldered into the top of the hot cylinder.

The schematic view of the measurement system is shown in Figure 10. To reveal possible faults in the efficiency measurements or in the engine itself and to map the behavior of the engine under various loads, all of the losses occurring in the system were determined so that a heat balance could be set up. Some of the mechanical losses could not be measured, but these can be found approximately by the difference from the heat balance. Heat was supplied by an electrical heater, which was designed and calculated according to the power required to achieve the design temperature. The power of the engine was determined by means of four measurements: the gas pressure in the engine, the speed of the engine, the temperature of the heater tubes, and the temperature of the cooler tubes. For this prototype engine, the working gas was supplied by an air compressor, and the charge pressure could be controlled by the pressure valve. These measurements were carried out, and the load on the engine was set to the desired value by a magnetic powder brake. The torque was measured by a dynamometer with loading accuracy of

**Figure 8.** The prototype Stirling engine.

0.03 Nm. A digital tachometer with 1-rpm accuracy was used to measure the speed of the engine. The temperature of the cooler tubes was not measured but was deduced from the inlet temperature of the cooling water and the flow rate of the water in the circuit cooler. Only one of the heater tubes was monitored by three NiCr–Ni-type thermocouples with 2.5°C accuracy because all of the 28 tubes were arranged axisymmetrically. The two pressure sensors were located at the inlet and the outlet of the regenerator.

The measuring instruments were calibrated before each test, and the calibration was checked at regular intervals during the test. All of the data required in the test, including speed, torque, pressure, and temperature, were recorded by the data collection instruments, and then, they were transferred to a PC for postprocessing.

4. EXPERIMENTAL RESULTS AND DISCUSSION

The test engine charged with air started to run at a 400°C hot end temperature with the help of a starter motor. For the resistance reason, the engine could not start operation

for a hot end temperature lower than 400°C. Experiments were conducted at three charge pressure levels, 5, 10, and 15 bar, and at heat source temperatures of 400, 500, and 600°C for each pressure. The engine speed was measured versus the torque of the engine, as the hot end temperature was constant at any desired value. The brake power output could be calculated by the following equation:

$$P = \frac{TN}{9550} \quad (14)$$



Figure 9. Heater tubes with connecting channel on the upper head.

where T is the torque of the Stirling engine and N is the rotation speed of the Stirling engine.

4.1. Impact of charge pressure on torque and power output

Variation of the engine torque and the power output, depending on the engine speed, is shown in Figures 11–13 for hot end temperatures of 400, 500, and 600°C. For all charge pressures, the torque of the engine reaches its maximum value when the running speed is at its minimum. The maximum torque increases with an increase in charge pressure. This variation in the trend of torque with engine speed is typical for a Stirling engine [34–38]. As is seen in these figures, when the charge pressure increases, the range of the running speed also increases. This phenomenon is attributed to the increase of inertia of the process materials, caused by the increasing charge pressure. With an increase in running speed, the power output first increases in the speed range from 200 to 300 rpm and then declines because of inadequate heat transfer and friction losses caused by high piston speed. At 600°C and 15 bar, the maximum shaft torque and power output reached 13.18 Nm at 200 rpm and 289 W at 250 rpm.

4.2. Impact of hot end temperature on torque and power output

The relationship between the engine torque and the power output in terms of the hot end temperature is illustrated in Figure 14. The graph is composed of the maximum torque and the power output values at 5, 10, and 15-bar charge pressures for each hot end temperature. It can be noticed that the torque and the power output increase as functions

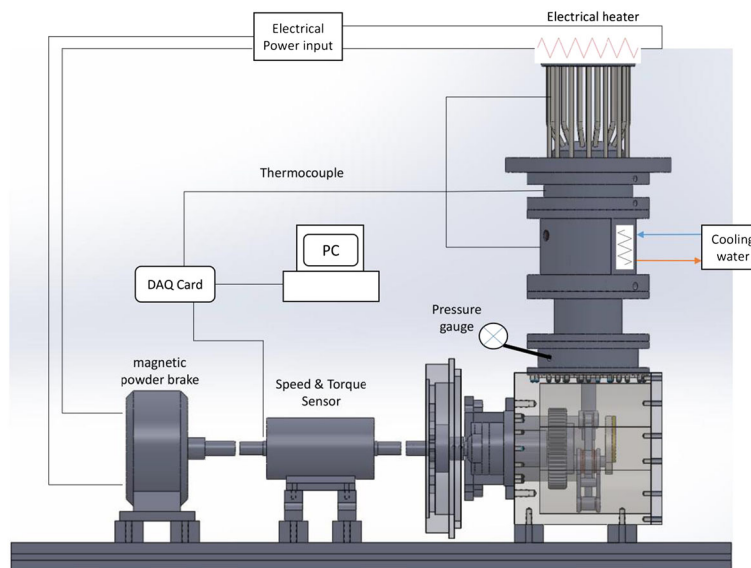


Figure 10. Schematic view of the measurement system.

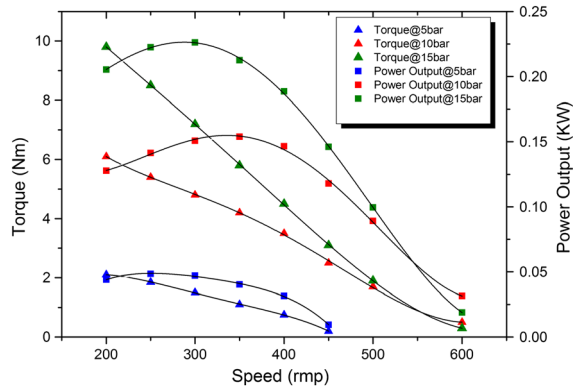


Figure 11. Variation of engine torque and brake power with engine speed at 400°C hot end temperature.

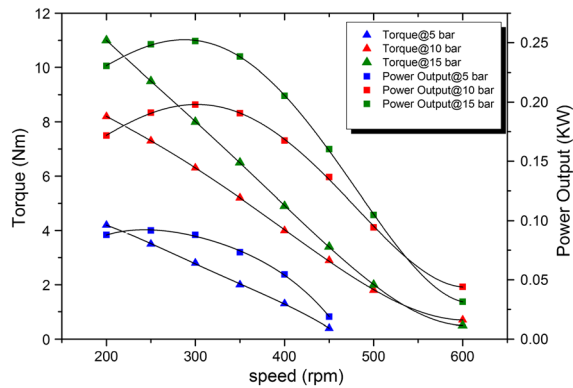


Figure 12. Variation of engine torque and brake power with engine speed at 500°C hot end temperature.

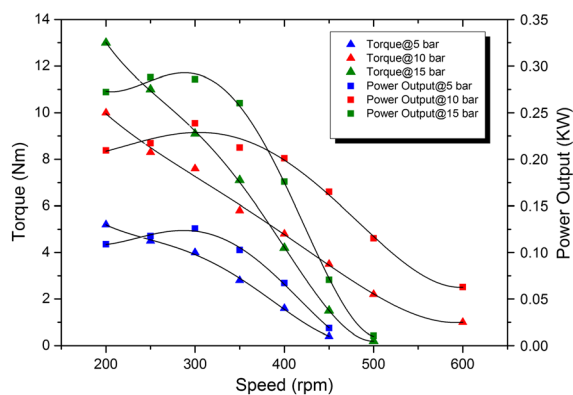


Figure 13. Variation of engine torque and brake power with engine speed at 600°C hot end temperature.

of hot end temperature for each charge pressure. Thus, the development of heat resistant material technology will lead to the promotion of Stirling engines.

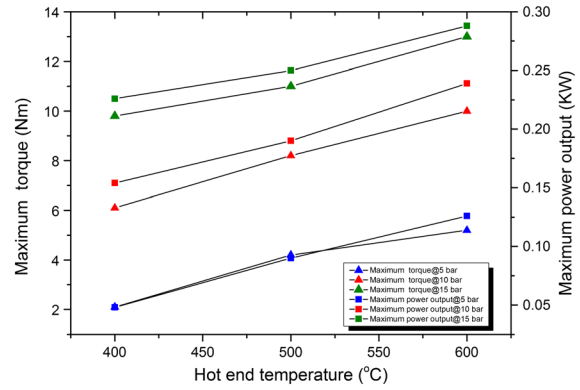


Figure 14. Variation of maximum power and maximum torque versus the hot end temperature.

5. CONCLUSIONS AND OUTLOOKS

In the present work, the rhombic drive mechanism for a Stirling engine has been analyzed. The development and experimental study of the engine prototype have been performed. This experimental test mainly studies the influence of the engine parameters (working temperature and charge pressure) on the engine performances (engine speed, torque, and power output).

By using an electrical heater as the heat source, the engine tests were conducted at various charge pressures. The starting temperature was measured as 400°C for a 5-bar charge pressure. The engine produced a maximum shaft power of 288 W at 600°C and 15-bar charge pressure. It has been shown that the power output of the Stirling engine increases with the initial charge pressure. Additionally, there appears to be a linear relationship between the hot end temperature and the power output. It can also be found that the power output of the Stirling engine increases with the engine speed until a maximum value, after which, it declines. This decline is because of the inadequate heat transfer and friction losses caused by high piston speed.

In this study, although the practical power output obtained during the tests does not reach the designed output, the results clearly demonstrate the typical variation trend of the Stirling engine. These preliminary test results are useful for the improvement of the engine redesign. It is expected to be necessary to improve the design of the sealing part of the Stirling engine and therefore to reduce the dead volumes of the heat exchangers. If the large dead volumes, the imperfect leakage, and the heat transmission are mitigated, the Stirling engine will promise better performance.

NOMENCLATURE

S	= stoke, cm
r	= crank radius, cm
e	= eccentricity of the drive mechanism, cm
L	= connecting rod length, cm
b	= distance between two yokes, cm

y	= displacement, cm
V	= volume, cm ³
A	= swept area, cm ²
θ	= crank angle, °
λ	= relative length of the connecting rod
ε	= relative length of the drive mechanism
λ_{xe}	= normalized length of expansion volume
λ_r	= normalized length of the regenerator
λ_{xc}	= normalized length of compression volume
α_{ff}	= normalized free-flow area
λ_h	= normalized hydraulic radius
Π_V	= volume porosity
N_T	= characteristic temperature ratio
N_{MA}	= characteristic Mach number
N_{SG}	= characteristic Stirling number
N_{TCR}	= characteristic thermal capacity ratio
N_F	= characteristic Fourier number for the regenerator
α	= volume phase angle
v	= dead space parameter
Ψ	= dimensionless pressure

Subscripts

d	= displacer
p	= piston
c	= compression
e	= expansion
1	= position at minimum compression volume
2	= position at minimum expansion volume
3	= position at maximum compression volume
4	= position at maximum expansion volume
max	= maximum value
min	= minimum value
clc	= clearance
sw	= swept

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